



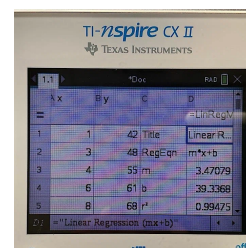
Question 1 [Maximum mark: 5]

A teacher records the number of hours, x , that each of eight students spent revising for a test, together with their test score, y , as a percentage.

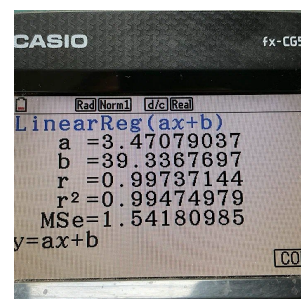
x (hours)	1	3	4	6	8	10	12	14
y (%)	42	48	55	61	68	74	81	87

(a) Find Pearson's product-moment correlation coefficient, r . [2]

Using ti-nspire CX ii After entering values in table: click→menu→4 statistics →1 stat calc→
3 Linear Regression



Using Casio CG-50 After entering values in two lists: click→F₂(calc)→F₃(REG)→F₁(x) → F₁(ax+b)



$$r \approx 0.997$$

(b) State the equation of the regression line of y on x . [1]

Using values of m and b from G.D.C.

$$y = 3.47x + 39.3$$

(c)(i) Use your regression line to estimate the test score of a student who revised for 9 hours.

(ii) Comment on the reliability of this estimate. [2]

$$(i) \quad y = 3.47(9) + 39.3$$

$$\therefore y = 70.5\%$$

(c)(ii) Reliability

Step 1: The observed x -values range from 1 to 14.

Step 2: Since $x = 9$ is within this range, this is interpolation. As $|r|$ is close to 1, the estimate is reasonably reliable.

[Maximum mark: 5]

(a) Find the coefficient of x^3 in the expansion of $(4 + 3x)^6$. [2]**(b)** Hence find the coefficient of x^3 in the expansion of $(2 - x)(4 + 3x)^6$.**(a)**

$$(4 + 3x)^6 = 6c_0 4^6 + 6c_1 (4)^5 (3x)^1 + 6c_2 (4)^4 (3x)^2 + 6c_3 (4)^3 (3x)^3 + \dots$$

$$\text{coefficient of } x^3 = 6c_3 (4)^2 (3)^3 = 34560$$

(b)

$$(2 - x) (4096 + 18432x + 34560x^2 + 34560x^3 + \dots)$$

$$\text{coefficient of } x^3 = 2 \times 34560 - 34560 = 34560$$

Question 3



[Maximum marks: 5]

The discrete random variable X has the probability distribution shown in the table, where $a, b \in \mathbb{Q}^+$

x	-10	2	5	8
$P(X=x)$	a	0.25	b	0.15

(a) Write down an equation relating a and b . [1]

$$a + 0.25 + b + 0.15 = 1$$

$$\therefore a + b = 1 - 0.40$$

$$\therefore a + b = 0.60 \quad \text{--- (i)}$$

The probability distribution shown in the table represents a fair game.

(b) Find the value of a and the value of b . [4]

The game is fair

$$\therefore E(x) = 0$$

$$\therefore -10a + 0.50 + 5b + 1.20 = 0$$

$$-10a + 5b = -1.70 \quad \text{--- (ii)}$$

From (i)

$$a = (0.60 - b)$$

From (ii)

$$-10(0.60 - b) + 5b = -1.70$$

$$-6 + 10b + 5b = -1.70$$

$$b = \frac{43}{150} \quad \therefore a = \frac{47}{150}$$

[Maximum mark: 8]

During a chemical reaction, the rate at which a compound is produced, in grams per minute, is modelled by

$$r(t) = 12te^{-3t}$$

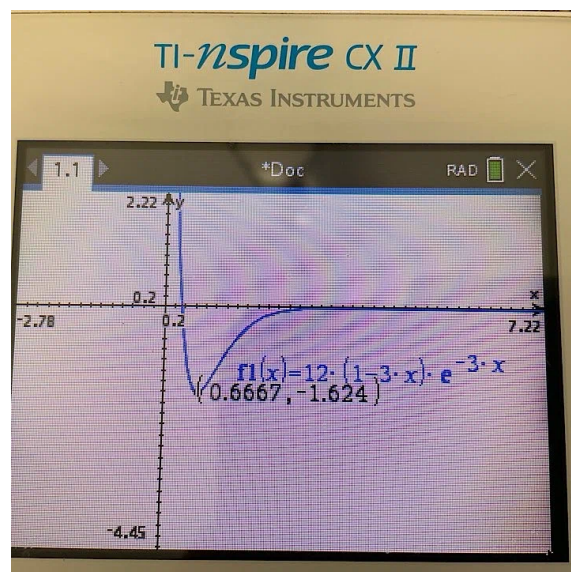
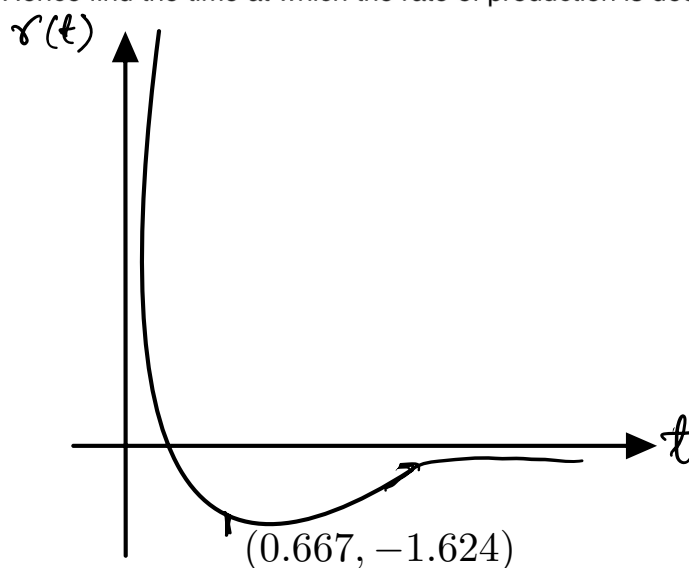
where t is the time in minutes after the start of the reaction.

(a) Find an expression for $r'(t)$. [3]

$$(uv)' = u'v + uv'$$

$$r'(t) = 12(1 \cdot e^{-3t} + t(-3e^{-3t})) = 12(1 - 3t)e^{-3t}$$

(b) Hence find the time at which the rate of production is decreasing most rapidly. [3]



$\therefore$ for $t \approx 0.667$

Rate is decreasing most rapidly

(using G.D.C.)

(c) Find the total mass of compound produced during the first 4 minutes. [2]

$$\text{Total mass} = \int_0^4 12te^{-3t} dt$$

$$\approx 1.333g \quad (\text{ using G.D.C.})$$

Question 5

[Maximum marks: 6]



The depth of water, d metres, in a harbour is modelled by $d(t) = a \sin(bt) + k$, $0 \leq t \leq 24$, where t is the time in hours after midnight and a, b, k are positive constants.

On a particular day the first high tide, of depth 8 m, occurs at $t = 3$, and the following low tide, of depth 3 m, occurs at $t = 9$.

(a) (i) Find the value of a . [1]

(ii) Find the value of k . [1]

$$(i) \quad a = \frac{\text{max} - \text{min}}{2} = \frac{8 - 3}{2} = 2.5$$

$$(ii) \quad k = \frac{\text{max} + \text{min}}{2} = \frac{8 + 3}{2} = 5.5$$

(b) Show that $b = \pi$ divided by twice the time interval between the high and low tides. Give the exact value of b . [1]

$$\text{From high tide to low tide time} = 9 - 3 = 6$$

$$\therefore \text{Period} = 2 \times 6 = 12$$

$$\therefore b = \frac{2\pi}{\text{Period}} = \frac{2\pi}{12} = \frac{\pi}{6}$$

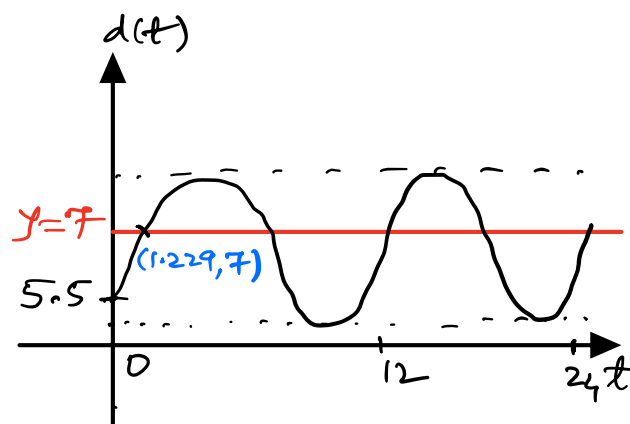
(c) Find the first time after midnight when the depth of water is exactly 7 metres. Give your answer to the nearest minute. [3]

$$d(t) = 2.5 \sin\left(\frac{\pi}{6}t\right) + 5.5$$

$$t = 1.229 = 1 + (0.229 \times 60)$$

$$\text{at } t \approx 1:14 \text{ a.m.}$$

depth will be 7m



Question 6

[Maximum marks: 7]

Consider the functions f and g defined by

$$f(x) = \frac{2x+1}{x-3}, \quad x \neq 3, \quad g(x) = x-1, \quad x \in \mathbb{R}.$$

(a) Find $f^{-1}(x)$.

[3]

$$y = \frac{2x+1}{x-3}$$

$$\therefore x = \frac{2y+1}{y-3}$$

$$x(y-3) = 2y+1$$

$$xy - 3x = 2y + 1$$

$$-3x - 1 = 2y - xy$$

$$-3x - 1 = y(2-x)$$

$$\text{or } y(x-2) = 3x+1$$

$$\therefore y = \frac{3x+1}{x-2}$$

$$\therefore f^{-1}(x) = \frac{3x+1}{x-2}$$

(b) Find an expression for $(f^{-1} \circ g)(x)$, giving your answer as a single fraction.

[2]

$$(f^{-1} \circ g)(x) = f^{-1}(g(x)) = f^{-1}(x-1)$$

$$= \frac{3(x-1)+1}{(x-1)-2} = \frac{3x-2}{x-3}$$

(c) Hence solve $(f^{-1} \circ g)(x) = -1.5$.

[2]

$$\frac{3x-2}{x-3} = -1.5$$

$$x = \frac{13}{9} \approx 1.44 \quad (\text{using G.D.C.})$$

SECTION B



Question 7

[Maximum marks: 13]

On 1 January 2021, a student deposits \$8,000.00 into a savings account that pays a nominal annual interest rate of 4.5%, compounded annually.

(a) Find the value of the account on 1 January 2025. Give your answer to two decimal places. [2]

$$N=4, \quad I=4.5\%, \quad PV=-8000, \quad Pmt=0, \quad FV=?$$

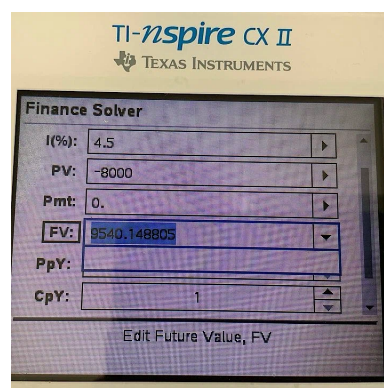
$$Ppy=1, \quad Cpy=1$$

Manually;

$$FV = \$9540.15$$

$$FV = PV(1+r)^4$$

$$= 8000(1.045)^4 \approx \$9540.15$$



(b) Find the number of complete years, after 1 January 2021, required for the value of the account to first exceed \$12,000.00. [3]

$$N=?, \quad I=4.5\%, \quad PV=-8000, \quad Pmt=0, \quad FV=12000$$

$$N=9.2 \approx 10 \text{ years}$$

Manually;

$$FV = PV(1+r)^n \Rightarrow 12000 = 8000(1+0.045)^n$$

$$n=9.21... \approx 10 \text{ years} \quad (\text{using G.D.C.})$$

The student also begins a separate monthly savings plan. They save \$15.00 in the first month, and in each month afterwards they save \$5.00 more than in the previous month.

(c) Find the amount saved in the 12th month. [2]

This is Arithmetic sequence, with

$$u_1 = 15, d = 5, n = 12$$

$$\therefore u_{12} = u_1 + 11d$$

$$= 15 + 11 \times 5$$

$$u_{12} = \$70$$

(d) Find the total amount saved during the first 12 months. [2]

$$u_1 = 15, d = 5, n = 12$$

$$\therefore S_n = \frac{n}{2} [2u_1 + (n - 1) d]$$

$$\therefore S_{12} = \frac{12}{2} [2 \times 15 + (12 - 1) 5]$$

$$\therefore S_{12} = \$510$$

(e) Find the number of years required for the total amount saved in the second plan to first exceed the value of the first savings account. Assume the comparison is made at the end of each complete year. [4]

First plan after m years,

$$FV = 8000 (1.045)^m$$

And after m complete years, the second plan has 12m monthly payments.

$$\therefore S_{12m} = \frac{12m}{2} [2u_1 + (12m - 1) d]$$

$$s_{12m} = 6m [30 + (12m - 1) 5]$$

$$8000 (1.045)^m = 6m [30 + (12m - 1) 5]$$

$$\therefore m = 5.06581 \dots \quad (\text{using G.D.C.})$$

For $m = 6$

$$1^{\text{st}} \text{ plan} \rightarrow FV = 8000 (1.045)^6 \approx \$10418.08$$

$$2^{\text{nd}} \text{ plan} \rightarrow S_{72} = \$13860 \quad (\text{using G.D.C.})$$

Therefore, in 6 years second plan total amount will exceed first plan

Question 8

[Maximum marks: 17]

A machine produces chocolate bars. The mass, M grams, of a bar is normally distributed with mean 50 and standard deviation 2.5.

- (a) Find the probability that a randomly selected chocolate bar weighs more than 53 grams.

[2]

$$X \sim N(50, 2.5^2)$$

$$\therefore P(X > 53) \approx 0.115 \quad (\text{using G.D.C.})$$

90% of the bars weigh more than m grams.

- (b) Find the value of m .

[2]

$$P(x > m) = 0.90$$

Using inverse normal

$$m \approx 46.8 \quad (\text{using G.D.C.})$$

The bars are packed into boxes of 20. A bar is classified as **underweight** if its mass is less than 46 grams.

- (c) Calculate the probability that a randomly chosen bar is underweight.

[2]

$$P(x < 46) \approx 0.0548 \quad (\text{using G.D.C.})$$

- (d) Find the probability that a box contains exactly 2 underweight bars.

[3]

Here we use Binomial Distribution

In a **binomial experiment**:

- there are a fixed number of **independent trials**
- there are only *two* possible results for each trial: **success** if some event occurs, or **failure** if the event does not occur
- the probability of success is the same for each trial.

If X is the number of successes in a binomial experiment with n trials, each with probability of success p , then X is a **binomial random variable**.

$$n = 20, p = 0.0548 \Rightarrow X \sim B(20, 0.0548)$$

$$p(x = 2) \approx 0.207$$

- (e) (i) Find the probability that a box contains at least 2 underweight bars.
(ii) Given that a box contains at least 2 underweight bars, find the probability that it contains exactly 2 underweight bars. [5]

(i) $P(x \geq 2) \approx 0.300$ (using G.D.C.)

(ii) Applying conditional Probability formula,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(x = 2|x \geq 2) &= \frac{P(x = 2|x \geq 2)}{P(x \geq 2)} = \frac{P(x = 2)}{P(x \geq 2)} \\ &= \frac{0.2068 \dots}{0.3004} \approx 0.689 \text{ (using G.D.C.)} \end{aligned}$$

The machine is recalibrated. The mass of a bar is still normally distributed with standard deviation 2.5 grams, but the mean is changed so that only 2% of bars are now underweight.

- (f) Find the new mean mass of a bar. [3]

A bar is classified as **underweight** if its mass is less than 46

Using standard normal Distribution $Z \sim (0, 1^2)$

$$\therefore P(Z < z) = 0.02$$

Using inverse normal, we get

$$z \approx -2.0537489 \text{ (using G.D.C.)}$$

Using standardization formula,

SL 4.12	Standardized normal variable	$z = \frac{x - \mu}{\sigma}$
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$$\therefore z = \frac{x - \mu}{\sigma}$$

$$-2.0537489 = \frac{46 - \mu}{2.5}$$

$$\therefore \mu = 51.13 \dots$$

$$\therefore \mu = 51.1 \text{ g}$$

Question 9

[Maximum marks: 14]



A piston moves back and forth along a straight line inside an engine. Its velocity, $v \text{ m.s}^{-1}$, at time t seconds is modelled by

$$v_1(t) = 5 \cos(t) - 2, \quad 0 \leq t \leq 8,$$

where t is measured in radians.

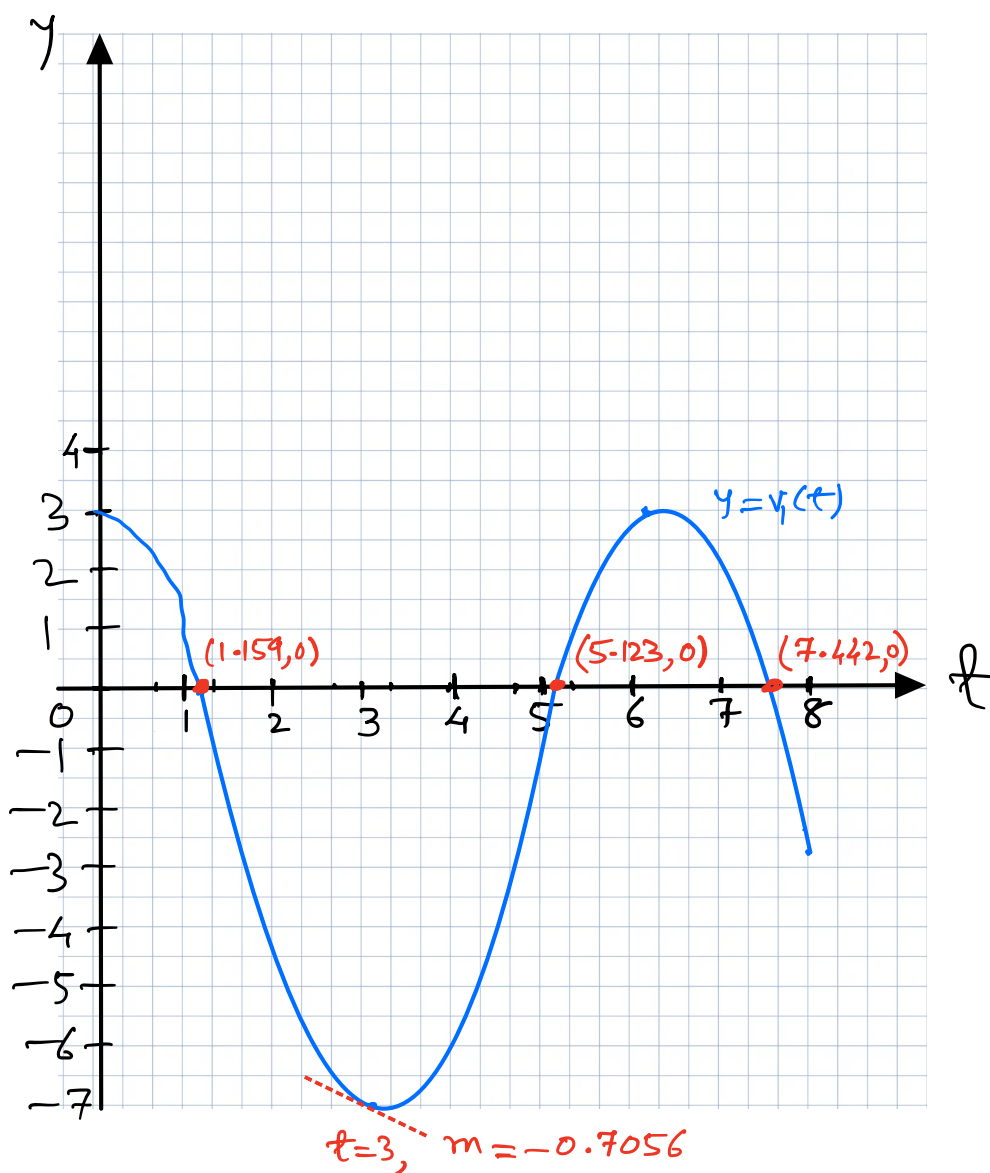
(a) Determine the velocity of the piston when $t = 0$.

[2]

$$v_1(0) = 5 \cos(0) - 2 = 5 - 2 = 3 \text{ m s}^{-1} \quad (\text{using G.D.C.})$$

(b) Find the values of t in the interval $0 \leq t \leq 8$ for which the piston is momentarily at rest.

[3]



Hence, from graph

$$t = 1.16 \text{ s}, t = 5.12 \text{ s}, t = 7.44 \text{ s}$$

(c) Find the acceleration of the piston when $t = 3$.

[2]



$$a(t) = v'(t)$$

From part(b) graph, on G.D.C., we get

$$v_1(t) = 5 \cos(t) - 2,$$

$$a(t) = v'_1(t) = -5 \sin t$$

$$\therefore a(3) = -5 \sin(3) = -0.7056 \dots$$

$$a(3) = v'(3) \approx -0.706 \text{ m s}^{-1}$$

The piston passes through the point O when $t = 0$.

(d) Find the total distance travelled by the piston during the first 4 seconds.

[3]

Using formula 5.9, we get

$$\left| \begin{array}{l} \text{5.9} \\ \text{Distance travelled from} \\ t_1 \text{ to } t_2 \end{array} \right| \quad \left| \text{distance} = \int_{t_1}^{t_2} |v(t)| dt \right|$$

$$\text{distance} = \int_{t_1}^{t_2} |v(t)| dt$$

$$= \int_0^4 |5 \cos(t) - 2| dt$$

$$\text{distance} = 16.3 \text{ m}$$

(using G.D.C.)

A second piston moves with a similar motion to the first piston, with velocity, $v \text{ m.s}^{-1}$, at time t seconds modelled by

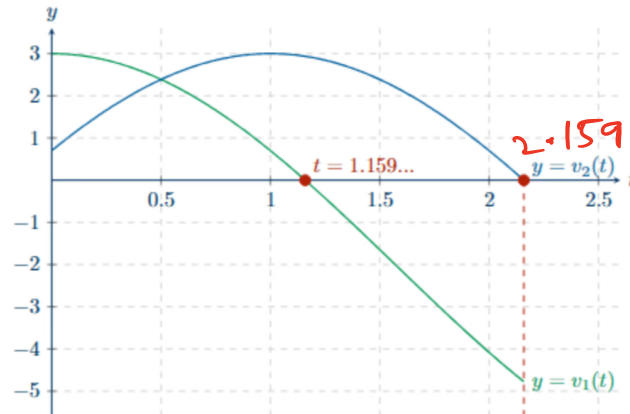
$$v_2(t) = 5 \cos(t - 1) - 2, \quad 0 \leq t \leq 8,$$

where t is measured in radians.

- (e) Determine how much further the first piston has travelled than the second piston, at the first time the second piston is momentarily at rest.

[4]

finding the zero of graph of second function $v_2(t)$, we get



$$t = 2.159279$$

$$\text{difference} = \int_{t_1}^{t_2} |v_1(t)| dt - \int_{t_1}^{t_2} |v_2(t)| dt$$

$$= \int_0^{2.159} |\cos(t) - 2| dt - \int_0^{2.159} |\cos(t - 1) - 2| dt$$

$$\text{difference} = 0.216 \text{ m}$$

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ON

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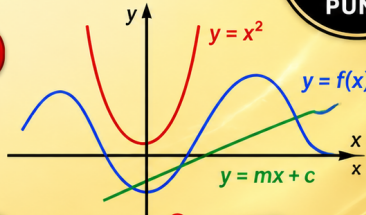
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